

Week 4 Office Hour Notes

4.0 Errata

9/22/24 11:20 am. Section 4.1 was edited. Other than changes in formatting, **Lemma 4.1.3** was added and proof of **Theorem 4.1.4**, i.e., the General Principle of Mathematical Induction, was added.

4.1 General Principle of Mathematical Induction

Definition 4.1.1. Given a universal set U , a set $B \subset U$ of "beginner terms", and a collection $\{f_1, \dots, f_m\}$ of n -ary functions $f_k : U \times \underbrace{\dots}_{n_k \text{ times}} \times U \rightarrow U$ with respect to B , we call S an

inductive set with respect to the beginner terms B and n -ary functions $\{f_1, \dots, f_m\}$ (or we call it an **inductive set** if the context of B and $\{f_1, \dots, f_m\}$ is clear) if the following conditions are satisfied:

1. (*Base Condition*) $B \subset S$; in other words, every $b \in B$ is also in S
2. (*Closure Condition*) For each n -ary function f_k , if $x_1, \dots, x_{n_k} \in S$, then $f_k(x_1, \dots, x_{n_k}) \in S$.

Examples 4.1.2.

1. Let's say we have a universal set U , distinct elements $a, b \in U$. Consider $B := \{a, b\}$ and the unary function $g : U \rightarrow U$ and the binary function $f : U \times U \rightarrow U$. Then define recursively

$$C_1 := B$$

$$C_{n+1} := C_n \cup \{f(x, y) : x, y \in C_n\} \cup \{g(x) : x \in C_n\},$$

%USE THIS DEFINITION AS AN EXAMPLE OF "DEFINITION BY RECURSION".

$$C_1 := \{a, b\},$$

$$C_2 := \{a, b, f(a, a), f(a, b), f(b, a), f(b, b), g(a), g(b)\},$$

$$C_3 := \{a, b, f(a, b), f(a, b), f(b, a), f(b, b), g(a), g(b), f(a, f(a, b)), f(b, f(a, b)), \dots\},$$

set $C_* := \bigcup_n C_n$. We find that C_* is an inductive set with respect to $\{a, b\}$ and f, g , which

we've discussed in the powerpoint is an inductive set.

2. Let's say we have the universal set $\mathbb{N} := \{1, 2, 3, \dots\}$, and the successor function $s : \mathbb{N} \rightarrow \mathbb{N}$ defined by $s(n) := n + 1$. Then $\mathbb{N}$ is an inductive set with respect to $B := \{1\}$ and the successor function s .

3. Let's say we have the universal set as all real numbers $\mathbb{R}$ and the successor function $s : \mathbb{R} \rightarrow \mathbb{R}$ defined by $s(x) := x + 1$ and a predecessor function $p : \mathbb{R} \rightarrow \mathbb{R}$ defined by $p(x) := x - 1$. Then the set $\mathbb{Z} := \{\dots, -2, -1, 0, 1, 2, \dots\}$ of integers is an inductive set with respect to $B := \{0\}$ and the successor and predecessor functions s, p .

4. Let's say we have the universal set U as the collection of strings of sentential logic symbols, and the negation function $\mathcal{E}_\neg : U \rightarrow U$ and the binary connective functions $\mathcal{E}_\vee, \dots : U \times U \rightarrow U$ defined as in the powerpoint in 1.1 back in 1.1. Then the set S of wff's is an inductive set with respect to B as the set of sentence symbols and the connective functions $\mathcal{E}_\neg, \mathcal{E}_\vee, \dots$ (note that this is so by definition because we define wff's to be closed given the collection of wff's $\alpha, \beta \in S$ under $\neg\alpha, \alpha \vee \beta, \dots$)

Recall the principles of mathematical induction we discussed in previous weeks:



Here are the two different principles of induction (in one of their possible phrasings) side-by-side:

Principle of Mathematical Induction: Let $S \subset \mathbb{N} := \{1, 2, 3, \dots\}$. If S has the following conditions:

1. (atomic condition) $1 \in S$,
2. (inductive step) If $n \in S$, then $n + 1 \in S$,

Then $S = \mathbb{N}$.

Principle of Structural Induction (on wff's): Let S be a collection of wff's. If S has the following conditions:

1. (atomic condition) $A_n \in S$, for every sentence symbol A_n ,
2. (inductive step) If $\alpha_1, \alpha_2 \in S$, then $\neg\alpha_1, \alpha_1 \wedge \alpha_2, \alpha_1 \vee \alpha_2, \alpha_1 \rightarrow \alpha_2, \alpha_1 \leftrightarrow \alpha_2 \in S$,

then S consists ALL wff's.

Note that these principles of mathematical induction is just the restatement of **Definition 4.1.1** applied to **part 2** and **4** of **Examples 4.1.2**, respectively.

Recall that C^* is the "smallest inductive set"

$$C^* := \bigcap_{S \text{ is inductive}} S, \quad (4.1)$$

which we show is in fact inductive.

Lemma 4.1.3. C^* is inductive.

Outline of Proof. Similar to how we show C^* , as defined in homework 4 problem 1, is inductive. Only difference is we verify the Closure Condition of Definition 4.1 by taking arbitrary n -ary function f_k and $x_1, \dots, x_{n_k} \in C^*$ and showing that $x_1, \dots, x_{n_k} \in C^*$. $\square$

Theorem 4.1.4. (*General Principle of Mathematical Induction*) Given a universal set U , a set $B \subset U$ of "beginner terms", and a collection $\{f_1, \dots, f_m\}$ of n -ary functions, define

$$\begin{aligned} C_1 &:= B, \\ C_{n+1} &:= C_n \cup \bigcup_{k=1}^m \{f_k(x_1, \dots, x_{n_k}) : x_1, \dots, x_{n_k} \in C_n\}, \quad (4.2) \\ C_* &:= \bigcup_{n=1}^{\infty} C_n. \end{aligned}$$

Then $C_* = C^*$.

Proof. Similar idea to showing $C_* = C^*$ in the case of $B := \{a, b\}$ and the functions as f, g in the powerpoint, by showing that $C_* \supset C^*$ and $C_* \subset C^*$.

First we need to show that $C_* \supset C^*$ and since C^* is the intersection of all inductive sets, it suffices to show that it's inductive. To show that it's inductive, we need to show the following:

1. (*Base Condition*) $B \subset C_*$, i.e. that it contains all base symbols (this follows immediately by definition of $C_1 := B$)

2. (*Closure Condition*) Given an n -ary function f_k and $x_1, \dots, x_{n_k} \in C_*$, we have $f_k(x_1, \dots, x_{n_k}) \in C_*$. This follows by a pretty straightforward analysis of $x_1, \dots, x_{n_k} \in C_*$, i.e. that $x_1, \dots, x_{n_k} \in C_n$ for $n \in \mathbb{N}$ sufficiently large, and the fact that $f_k(x_1, \dots, x_{n_k}) \in C_{n+1} \subset C_*$ by definition.

Next, we show that $C_* \subset C^*$, which follows by showing by induction on n that $C_n \subset S$, for all

$n \in \mathbb{N}$, for an arbitrary inductive set $S \subset U$ with respect to B and $\{f_1, \dots, f_m\}$. Such verification will be left to the reader. $\square$

4.2 Definition by Recursion

%GIVE PREAMBLE AND CORRECT MISTAKES

Definition by Recursion on the Natural Numbers $\mathbb{N}$. Given a function $f : \mathbb{N} \rightarrow \mathbb{N}$, there exists a unique function $g : \mathbb{N} \rightarrow \mathbb{N}$ defined by $g(1) = a$, for arbitrary $a \in \mathbb{N}$ and $g(s(n)) = g(n + 1) = f(n)$.

The idea is that from this definition, we can use inductive iteration to define everything that we need by going through the natural numbers one by one (using the successor function) and defining

$$\begin{aligned} g(1) &:= a \\ g(2) &:= f(a) \\ g(3) &:= f(f(a)) \\ g(n) &:= f(f(\underbrace{\quad \quad}_{n \text{ times}} (1) \dots)) \end{aligned}$$

Given a set A define the notation:

$$A^n := A \times A \underbrace{\quad \quad}_{n \text{ times}} \times A = \{(a_1, \dots, a_n) : a_1, \dots, a_n \in A\}$$

Now we'll take an arbitrary universal set U , an arbitrary set B of beginner terms, and an arbitrary set $\{f_1, \dots, f_m\}$ of n -ary functions, and define "Definition by Recursion" more generally. First we need to define a condition to ensure that definition by recursion works.

Definition 4.2.2. We consider a collection of n -ary functions $\{f_1, \dots, f_m\}$ a **symbollic collection of functions with respect to B**

1. given $b_1, \dots, b_{n_k} \in B$ and $1 \leq k \leq m$, we have $f_k(b_1, \dots, b_{n_k}) \notin B$.
2. given $x_1, \dots, x_{n_k} \in U$, we have $f_k(x_1, \dots, x_{n_k}) \neq f_j(x_1', \dots, x_{n_j}')$, for all $1 \leq j \neq k \leq m$ and $x_1', \dots, x_{n_j}' \in U$.

In other words, the image sets

$$f_k[U^{n_k}] := \{f(x_1, \dots, x_{n_k}) : x_1, \dots, x_{n_k} \in U\},$$

for each $1 \leq k \leq m$ pairwise disjoint from each other, i.e.,

$$f_k[U^{n_k}] \cap f_j[U^{n_j}] = \emptyset, \text{ for } 1 \leq j \neq k \leq m.$$

Theorem 4.2.3. (*Definition by Recursion*) Assume that $\{f_1, \dots, f_m\}$ is a symbolic set of n -ary injective functions with respect to B , and define C^* as in (4.1), and let A be some set.

Given a function $h : B \rightarrow A$ and a collection of n -ary functions $F_k : (C^*)^{n_k} \rightarrow A$, we can define the function $\bar{h} : C^* \rightarrow U$ by

$$\begin{aligned} \bar{h}(x) &= h(x) \text{ for } x \in B \\ \bar{h}(f_k(x_1, \dots, x_{n_k})) &= F_k(\bar{h}(x_1), \bar{h}(x_2), \dots, \bar{h}(x_{n_k})), \text{ for each } 1 \leq k \leq m, \end{aligned} \quad (4.3)$$

that extends h . Moreover, this function $\bar{h}$ is uniquely determined.

Examples 4.2.4.

1. define $h : \{1\} \rightarrow \mathbb{N}$ to be $h(1) = 1$, define $F : \mathbb{N} \rightarrow \mathbb{N}$ by $F(n) := 1 - n$. Then we can define $\bar{h}$, which as discussed in the discussion forum ends up being

$$\bar{h}(n) = \begin{cases} 1 & \text{for odd numbers} \\ 0 & \text{for even numbers} \end{cases}$$

2.

%TALK ABOUT DEFINING C_n BY RECURSION

3. If we have the valuation mapping $v : S \rightarrow S$ for a set S of sentence symbols, as discussed in week 2, we can define the truth assignment mapping $\bar{v} : \bar{S} \rightarrow \{T, F\}$ by

$$\begin{aligned} \bar{v}(A) &:= v(A), \text{ for all sentence symbols } A \in S \\ \bar{v}(\mathcal{E}_{\neg}(\alpha)) &= \neg \bar{v}(\alpha) \\ \bar{v}(\mathcal{E}_{\vee}(\alpha, \beta)) &= \bar{v}(\alpha) \vee \bar{v}(\beta) \\ \vdots & \qquad \qquad \qquad \vdots \end{aligned}$$

where $\neg : \{T, F\} \rightarrow \{T, F\}$, $\vee : \{T, F\} \times \{T, F\} \rightarrow \{T, F\}$ defined as one would with truth

tables (I plan to have some extracurricular notes on boolean algebras to explain more exactly what I mean).

Note that this is a special case of definition by recursion.

Lemma 4.2.5. For each $n \in \mathbb{N}$, let C_n be defined as in (4.2). if $\{f_1, \dots, f_m\}$ is a symbolic set of injective n -ary functions with respect to B , then for each $n \in \mathbb{N}$ we have

$$f_k(x_1, \dots, x_{n_k}) \notin C_{n+1}, \text{ for every } 1 \leq k \leq m \text{ and } (x_1, \dots, x_{n_k}) \in (C_{n+1})^{n_k} \setminus (C_n)^{n_k}, \quad (4.4)$$

and the satisfaction of **condition 2**, as described in **Definition 4.2.2**.

Proof. Left as an exercise (albeit, a rather hard one) for the reader. As a hint, we prove (4.4) by induction on n , where in the inductive step, we note that (4.3) holds by inductive hypotheses, we use **condition 2** of $\{f_1, \dots, f_m\}$ being symbolic with respect to B , along with the fact that by (4.2) we have

$$C_{n+2} = C_{n+1} \cup \bigcup_{k=1}^m f_k[(C_{n+1})^{n_k} \setminus (C_n)^{n_k}],$$

and the fact that f_k being injective for each $1 \leq k \leq m$ implies that the condition

$$(x_1, \dots, x_{n_k}) \neq (x_1', \dots, x_{n_k}') \in (C_{n+1})^{n_k} \implies f_k(x_1, \dots, x_{n_k}) \neq f_k(x_1', \dots, x_{n_k}'),$$

holds. $\square$

Proof of Theorem 4.2.3. To show existence, first note by **Theorem 4.1.4**, i.e. the *General Principle of Mathematical Induction*, that we have

$$C^* = C_* = \bigcup_{n=1}^{\infty} C_n,$$

where $\{C_n\}_{n \in \mathbb{N}}$, C_* are as defined in (4.2), it shall suffice to show that for every $n \in \mathbb{N}$, we can define $h_n : C_n \rightarrow A$, with $h_1 := h$, that extends $h_q : C_q \rightarrow A$, for each $1 \leq q < n$, that satisfies (4.3), for all $1 \leq k \leq m$ and all $x_1, \dots, x_{n_k} \in C_{n-1}$ (with $C_0 := \emptyset$). This allows us to define $\bar{h} : C^* \rightarrow A$ by $\bar{h}(x) := h_n(x)$, for $n \in \mathbb{N}$ such that $x \in C_n$, which is a well-defined function since the value $h_n(x)$ is uniquely determined since $h_{n'} : C_{n'} \rightarrow A$ is extended by h_n for every $1 \leq n < n'$ and an extension of h_n , for every $n' \geq n$.

We shall establish existence of such h_n , for each $n \in \mathbb{N}$, by induction on n . For the base case $n = 1$, we have $h_1 := h$ and we are done since $1 \leq q < n$ in this case does not exist and so the desired extension property holds in the base case vacuously.

For the inductive step, assume such h_n exists extending h_q , for each $1 \leq q \leq n$, that satisfies (4.3). Define $h_{n+1} : C_{n+1} \rightarrow A$ by

$$h_{n+1}(x) := \begin{cases} h_n(x) & \text{if } x \in C_n, \\ F_k(h_n(x_1), \dots, h_n(x_{n_k})) & \text{if } x \notin C_n \text{ and } x = f_k(x_1, \dots, x_{n_k}), \text{ for some } \\ & 1 \leq k \leq m \text{ and } x_1, \dots, x_{n_k} \in C_n. \end{cases} \quad (4.5)$$

First, note that h_{n+1} is well-defined, since by (4.2), **Lemma 4.2.5**, and **condition 2** of **Definition 4.2.2**, we have

$$C_{n+1} = C_n \cup \bigcup_{k=1}^m f_k[(C_n)^{n_k} \setminus (C_{n-1})^{n_k}], \text{ where } C_0 := \emptyset$$

with $\{C_n, f_1[(C_n)^{n_1} \setminus (C_{n-1})^{n_1}], \dots, f_m[(C_n)^{n_m} \setminus (C_{n-1})^{n_m}]\}$ all pairwise disjoint. As a result, we find if $x \in C_n$, then $h_{n+1}(x) = h_n(x)$, which is uniquely determined by inductive hypothesis. Or in the case where $x \notin C_n$, we have $x = f_k(x_1, \dots, x_{n_k})$, for some unique $1 \leq k \leq m$, and $(x_1, \dots, x_{n_k}) \in (C_n)^{n_k} \setminus (C_{n-1})^{n_k}$, in which case we have

$$h_{n+1}(x) = F_k(h_n(x_1), \dots, h_n(x_{n_k})),$$

hence given $y \in A$ such that $h_{n+1} : x \mapsto y$, we have

$$y = F_k(h_n(x_1'), \dots, h_n(x_{n_k}')), \text{ where } x = f_k(x_1', \dots, x_{n_k}'),$$

so by injectivity of f_k , we have

$$\begin{aligned} f_k(x_1', \dots, x_{n_k}') = x = f_k(x_1, \dots, x_{n_k}) &\implies (x_1', \dots, x_{n_k}') = (x_1, \dots, x_{n_k}) \\ &\implies y = F_k(h_n(x_1'), \dots, h_n(x_{n_k}')) \\ &= F_k(h_n(x_1), \dots, h_n(x_{n_k})) \\ &= h_{n+1}(x), \end{aligned}$$

and we conclude that the value $h_{n+1}(x)$ is uniquely determined.

Next, we find that h_{n+1} extends h_q for every $1 \leq q < n + 1$ since in the case where $q = n$, we have h_{n+1} extending h_n by (4.4), and in the case where $q < n$, we find h_n extends h_q , hence h_{n+1} as an extension of h_n also extends h_q , by inductive hypothesis. This completes the inductive step of our proof of existence of h_n .

To show uniqueness, it suffices to show that if $h' : C^* \rightarrow A$ is a function extending h such that (4.3) holds, then given $x \in C^*$, we have $h'(x) = \bar{h}(x)$. This will be left as an exercise to the reader. $\square$

4.3 Questions